Deep Space Return Home Scenario

Group 4 - Olivia Guzik, Solomon Jones, Kathleen Melonashi, Adam Shaar

As Alex delves deeper into the mysteries of the Infinity Edge Explorer, he gradually discovers how time is of the essence in his return home. Due to a lack of maintenance and years of disarray, the vessel's systems are deteriorating. To fix the different systems around the ship, Alex must overcome a series of tasks and challenges to safely navigate the ship back to the home planet.

Scenario "Return Home"

Some systems need to be fixed throughout the ship before Alex can safely return. Firstly, the lights are out, and there is only one way to see the ship: with a flashlight. To fix the lights, Alex will get subtle hints about where the button will enable the electricity.

1) *Turn on the lights:*
   Alex finds the button to turn on the lights but notices issues with the electric panel responsible for the ship's lighting. Wires are hanging loose, the circuit board is flickering with sparks, and warning lights are flashing around the panel. To restore the main lights of the ship, Alex must delve into the intricate configurations of the circuit and troubleshoot the malfunctioning components.

Alex learns that there is a moving robot on the ship, but it seems more hostile than helpful. To get more helpful clues on how to return home, Alex must learn how to repair the robot and fix its internal system.

2) *Repair the Robot:*
   A robot is seen sparking with broken joints and emitting odd noises. It randomly crashes into walls and acts erratically around Alex. Alex must find broken robot parts around the ship to repair the robot. Once the physical repairs are complete, Alex encounters a "Code Debugging" challenge. Alex must identify logical errors and correct the code using basic programming principles like loops, conditionals, and function calls.

Alex starts to feel lightheaded when he learns that the ship's oxygen replenishment system is broken. The oxygen system is critical, relying on filters, compressors, and chemical reactors to remove carbon dioxide and replenish oxygen. However, a critical software glitch has disrupted the system's automation, requiring Alex to recalibrate and restart the process manually.

3) *Fix the Oxygen:*

Alex needs to replace the oxygen tank on the ship's filtration system. He must explore the ship's different rooms, looking for the tank. Once found and replaced, Alex must fix the code for the automation system to get the oxygen filter working again. He will have to turn dials on an interface to fix the system. Alex can find hints on how to do so properly.

As Alex explores the ship, he finds data logs and encrypted messages describing the ship's original mission. To uncover more information, Alex must decrypt it and piece it together to return home.

4) *Piece together the hidden code:*
   Decrypt encrypted data logs and messages scattered throughout the ship. These data logs are stored in notebooks and computers in different rooms throughout the ship. To decrypt the messages, Alex needs to activate the ship's robot. After decrypting the messages, Alex must put together the pieces to understand the mission and how to return home.

Now that Alex has a sustained method of survival and a better understanding of why and where the Explorer has ventured off, he needs to connect back to the home planet to inform them of his situation and an evacuation method.

5) Contact the Mother Planet

Alex must rebuild the communication array and adjust the position of the ship's satellites to send an encrypted distress signal to his home planet so that it can perform a rescue mission. He will also need to calculate the correct angles between the ship and Earth so that the message is properly intercepted. Once the signal is properly aligned, Alex must send an encrypted distress signal located somewhere on the ship.

Once Alex completes the final task, his ship can safely self-navigate itself home, queuing the game's end screen.